

THE EIGENVALUES AND THE EIGENVECTORS

MATRIX DIAGONALIZATION

A. prof. Elsayed Mahdy
EAEA

THE EIGENVALUES AND THE EIGENVECTORS

Eigenvalues and Eigenvectors.

Let A be an $n \times n$ matrix. A real or complex number λ is called an **eigenvalue of A** if, for some nonzero $n \times 1$ matrix X ,

$$AX = \lambda X$$

Any nonzero $n \times 1$ matrix X satisfying this equation for some number λ is called an **eigenvector of A associated with the eigenvalue λ** .

Theorem: Let A be an $n \times n$ matrix. Then

1. λ is an eigenvalue of A if and only if $|A - \lambda I| = 0$
2. If λ is an eigenvalue of A , any nontrivial solution of $(A - \lambda I)X = 0$ is an eigenvectors of A associated with λ .

Notes:

1. We therefore find the eigenvalues of A by solving the equation, $|A - \lambda I| = 0$.
2. Given A , we expand this determinant, obtaining a polynomial of degree n in λ . This polynomial is called **the characteristic polynomial of A** , and its roots are the eigenvalues of A .
The equation $|A - \lambda I| = 0$ is **the characteristic equation of A** .
3. Associated with each eigenvalue λ the nontrivial solutions of $(A - \lambda I)X = 0$ are the eigenvectors of A

Example (1): Find the eigenvalues and eigenvalues vectors of the matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$

Solution

To Find the eigenvalues of the matrix A , we solve the characteristic equation: $|A - \lambda I| = 0$

Therefore, $|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 1 \\ -2 & 4 - \lambda \end{vmatrix} = 0$

$$(1 - \lambda)(4 - \lambda) + 2 = 0 \quad \Rightarrow \quad \lambda^2 - 5\lambda + 6 = 0 \quad \Rightarrow \quad (\lambda - 2)(\lambda - 3) = 0$$

Therefore, the eigenvalues of A are: $\lambda = 2, 3$

Now, we want to find the all non-zero eigenvectors $X_{1,2}$ of A corresponding to the eigenvalues $\lambda_{1,2}$ that satisfy the equation,

$$(A - \lambda I)X = 0 \quad \rightarrow \quad (1)$$

the last equation yields the following homogeneous system:

$$\begin{cases} (1 - \lambda)x_1 + x_2 = 0 \\ -2x_1 + (4 - \lambda)x_2 = 0 \end{cases} \rightarrow (2)$$

To find the eigenvector $X_{\lambda=2}$, put $\lambda = 2$, in system (2), then we get,

$$\left. \begin{array}{l} -x_1 + x_2 = 0 \\ -2x_1 + 2x_2 = 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} -x_1 + x_2 = 0 \\ -x_1 + x_2 = 0 \end{array} \right\} \Rightarrow x_1 = x_2 \quad \text{Therefore, put,} \\ x_1 = x_2 = r \quad \text{where } r \text{ is an arbitrary non-zero real number, then}$$

$$\therefore X_{\lambda=2} = \begin{bmatrix} r \\ r \end{bmatrix} \quad \text{where } r \text{ is an arbitrary non-zero real number.}$$

Similarly, to find the eigenvector $X_{\lambda=3}$, put $\lambda = 3$, in system (2), then we get;

$$\left. \begin{array}{l} -2x_1 + x_2 = 0 \\ -2x_1 + x_2 = 0 \end{array} \right\} \Rightarrow -2x_1 + x_2 = 0 \Rightarrow 2x_1 = x_2$$

$$\text{put, } x_1 = r \Rightarrow x_2 = 2r \text{ therefore, } X_{\lambda=3} = \begin{bmatrix} r \\ 2r \end{bmatrix}$$

Example (2): Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -5 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Solution

The characteristic equation of A is: $|A - \lambda I| = 0$ therefore,

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & -1 & 1 \\ -5 & -2 - \lambda & 1 \\ 0 & 0 & 1 - \lambda \end{vmatrix} = 0,$$

Expand this determinant from the last row (**it contains two zeros**), then

$$(1 - \lambda)\{(2 - \lambda)(-2 - \lambda) - 5\} = 0 \Rightarrow (1 - \lambda)\{(2 - \lambda)(-2 - \lambda) - 5\} = 0 \\ (1 - \lambda)(\lambda^2 - 9) = 0 \Rightarrow (1 - \lambda)(\lambda - 3)(\lambda + 3) = 0$$

Therefore, are the eigenvalues of A are, $\lambda = 1, 3, -3$

Now, we want to find the all non-zero eigenvectors $X_{1,2,3} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ of A corresponding to $\lambda_{1,2,3}$

that satisfy the equation,

$$(A - \lambda I)X = 0 \rightarrow (1)$$

Equation (1) yields the following homogeneous system of equations:

$$\left. \begin{array}{l} (2 - \lambda)x_1 - x_2 + x_3 = 0 \\ -5x_1 - (2 + \lambda)x_2 + x_3 = 0 \\ (1 - \lambda)x_3 = 0 \end{array} \right\} \rightarrow (2)$$

To find the eigenvector $X_{\lambda=1}$, put $\lambda = 1$, in system (2), we get,

$$\left. \begin{array}{l} x_1 - x_2 + x_3 = 0 \\ -5x_1 - 3x_2 + x_3 = 0 \end{array} \right\} \Rightarrow 6x_1 + 2x_2 = 0 \Rightarrow x_2 = -3x_1$$

Let, $x_1 = r$ then we get $x_2 = -3r$ and $x_3 = -4r$. Therefore, $X_{\lambda=1} = \begin{bmatrix} r \\ -3r \\ -4r \end{bmatrix}$

To find the eigenvector $X_{\lambda=3}$, put $\lambda = 3$, in system (2), then we get;

$$\left. \begin{array}{l} -x_1 - x_2 + x_3 = 0 \\ -5x_1 - 5x_2 + x_3 = 0 \\ -2x_3 = 0 \end{array} \right\} \Rightarrow x_3 = 0 \Rightarrow x_1 + x_2 = 0 \Rightarrow x_1 = -x_2$$

Let $x_1 = r$, and $x_2 = -r$, $x_3 = 0$. Therefore, the eigenvector $X_{\lambda=3} = \begin{bmatrix} r \\ -r \\ 0 \end{bmatrix}$

To find the eigenvector $X_{\lambda=-3}$, put, $\lambda = -3$, in system (2), then we get

$$\left. \begin{array}{l} 5x_1 - x_2 + x_3 = 0 \\ -5x_1 + x_2 + x_3 = 0 \\ 4x_3 = 0 \end{array} \right\} \Rightarrow x_3 = 0 \Rightarrow 5x_1 - x_2 = 0 \Rightarrow x_2 = 5x_1$$

Let $x_1 = r$ and, $x_2 = 5r$, $x_3 = 0$. Therefore, the eigenvector: $X_{\lambda=-3} = \begin{bmatrix} r \\ 5r \\ 0 \end{bmatrix}$

Properties of Eigenvalues:

1. If α is nonzero scalar and λ is an eigenvalue of a matrix A , then $\alpha\lambda$ is an eigenvalue of the matrix αA .
2. If λ is an eigenvalue of a matrix A , then $1/\lambda$ is an eigenvalue of the matrix A^{-1} .
3. If λ is an eigenvalue of a matrix A , then λ is an eigenvalue of the matrix A^T .
4. The sum of eigenvalues of a matrix is equal to the sum of its main diagonal elements.
5. The eigenvalues of a triangular matrix are the main diagonal elements of the matrix.
6. The value of determinate of a square matrix equals the product of its eigenvalues.
7. A square matrix is singular if and only if it has an eigenvalue equal zero.

Example (3): If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ has eigenvalues **2, 3** and **k**.

i) Find the value of **k** and $|A|$.

ii) Find the eigenvalues of the matrices: $2A$, A^T , A^2 , A^{-1} .

iii) Find the eigenvectors of A ,

Solution

i) Since, the sum of eigenvalues of A is equal to the sum of main diagonal elements (trace (A))

therefore, $2 + 3 + k = 3 + 5 + 3 \Rightarrow k = 6$ then, eigenvalues $\lambda = 2, 3, 6$ and $|A| = 36$.

ii) the eigenvalues of the matrix $2A$ are: $\lambda = 4, 6, 12$

the eigenvalues of the matrix A^T are: $\lambda = 2, 3, 6$

the eigenvalues of the matrix A^2 are: $\lambda = 4, 9, 36$

the eigenvalues of the matrix A^{-1} are: $\lambda = \frac{1}{2}, \frac{1}{3}, \frac{1}{6}$

iii) Now, we want to find the all non-zero eigenvectors, $X_{2,3,6} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ of A corresponding to $\lambda_{2,3,6}$

that satisfy the equation: $(A - \lambda I)X = 0 \rightarrow (1)$

Equation (1) yields the following homogeneous system of equation:

$$\left. \begin{aligned} (3 - \lambda)x_1 - x_2 + x_3 &= 0 \\ -x_1 - (5 - \lambda)x_2 + x_3 &= 0 \\ x_1 - x_2 + (3 - \lambda)x_3 &= 0 \end{aligned} \right\} \rightarrow (2)$$

To find the eigenvector $X_{\lambda=2}$, put $\lambda = 2$, in system (2), then we get:

$$\left. \begin{aligned} x_1 - x_2 + x_3 &= 0 \\ -x_1 - 3x_2 - x_3 &= 0 \\ x_1 - x_2 + x_3 &= 0 \end{aligned} \right\} \Rightarrow \left. \begin{aligned} x_1 - x_2 + x_3 &= 0 \\ -x_1 - 3x_2 - x_3 &= 0 \end{aligned} \right\} \Rightarrow -4x_2 = 0 \Rightarrow x_2 = 0$$

then, $x_1 + x_3 = 0 \Rightarrow x_1 = -x_3$. Let $x_1 = r \Rightarrow x_3 = -r$. Therefore, $X_{\lambda=2} = \begin{bmatrix} r \\ 0 \\ -r \end{bmatrix}$

where r is an arbitrary non-zero real number.

To find the eigenvector $X_{\lambda=3}$, put $\lambda = 3$, in system (2), then we get;

$$\left. \begin{aligned} 0 - x_2 + x_3 &= 0 \\ -x_1 - 2x_2 - x_3 &= 0 \\ x_1 - x_2 - 0 &= 0 \end{aligned} \right\} \Rightarrow x_1 = x_2 \text{ \& } x_2 = x_3 \Rightarrow x_1 = x_2 = x_3$$

Let $x_1 = r$ we get: $x_1 = x_2 = x_3 = r$. Therefore, $X_{\lambda=3} = \begin{bmatrix} r \\ r \\ r \end{bmatrix}$

To find the eigenvector $X_{\lambda=6}$, put $\lambda = 6$, in system (2), then we get,

$$\left. \begin{array}{l} -3x_1 - x_2 + x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \\ x_1 - x_2 - 3x_3 = 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} x_1 + x_2 + x_3 = 0 \\ -3x_1 - x_2 + x_3 = 0 \\ x_1 - x_2 - 3x_3 = 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} x_1 + x_2 + x_3 = 0 \\ x_2 + 2x_3 = 0 \end{array} \right\}$$

Let $x_3 = r$ we get: $x_2 = -2r$ and $x_1 = r$ Therefore, $X_{\lambda=2} = \begin{bmatrix} r \\ -2r \\ r \end{bmatrix}$.

Cayley - Hamilton Theorem.

Theorem: Every square matrix satisfies its characteristic equation. i.e. if

$$|A - \lambda I| = a_0 + a_1\lambda + a_2\lambda^2 + \dots + a_n\lambda^n = 0$$

is the characteristic equation of a square matrix A , then

$$a_0I + a_1A + a_2A^2 + \dots + a_nA^n = 0$$

where λ has been replaced by A .

Note: By the Cayley-Hamilton theorem, we have the relation

$$a_0I + a_1A + a_2A^2 + \dots + a_nA^n = 0 \quad ; \quad |A| \neq 0$$

we obtain, $A^{-1} = -\left(\frac{a_1}{a_0}\right)I - \left(\frac{a_2}{a_0}\right)A - \left(\frac{a_3}{a_0}\right)A^2 - \dots - \left(\frac{a_n}{a_0}\right)A^{n-1}$

Example (1): If $A = \begin{bmatrix} 2 & 1 \\ -4 & -3 \end{bmatrix}$, verify Cayley-Hamilton theorem for A and then find A^{-1} .

Solution

The characteristic equation of A is : $|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 \\ -4 & -3-\lambda \end{vmatrix} = 0$

Therefore, the characteristic polynomial is; $\lambda^2 + \lambda - 2 = 0$

Cayley-Hamilton theorem states that; *square matrix satisfies its characteristic equation.*

Therefore, we must show that $A^2 + A - 2I = 0$??????

Substituting in left hand side of Cayley-Hamilton theorem, we get,

$$L.H.S = A^2 + A - 2I = \begin{bmatrix} 0 & -1 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ -4 & -3 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

We use Cayley-Hamilton theorem to find the inverse of A as follow:

$$\therefore A^2 + A - 2I = 0 \quad \Rightarrow \quad 2I = A^2 + A \quad \Rightarrow \quad I = \frac{1}{2}(A + I). A = A^{-1}. A$$

$$\therefore A^{-1} = \frac{1}{2}(A + I) = \frac{1}{2} \left\{ \begin{bmatrix} 2 & 1 \\ -4 & -3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \frac{1}{2} \begin{bmatrix} 3 & 1 \\ -4 & -2 \end{bmatrix} = \begin{bmatrix} 3/2 & 1/2 \\ -2 & -1 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 3/2 & 1/2 \\ -2 & -1 \end{bmatrix}$$

Example (2): If $A = \begin{bmatrix} 1 & 4 & 3 \\ 0 & c & 5 \\ 0 & 0 & 7 \end{bmatrix}$ and the $\text{trace}(A) = 6$.

- i) Find the value of c .
- ii) Find the eigenvalues of the matrices A , $3A$, A^T , A^2 , A^{-1} .
- iii) Verify Cayley-Hamilton theorem for the matrix A
- iv) Use Cayley-Hamilton theorem to find A^{-1}

Solution

i) $\therefore \text{trace}(A) = \text{sum of diagonal elements}$, therefore, $6 = 1 + c + 7 \Rightarrow c = -2$

Since, the matrix A is an upper triangular matrix, therefore from the properties of the eigenvalues of A we know that:

- ii) the eigenvalues of the matrix A are: $\lambda = 1, -2, 7$
the eigenvalues of the matrix $3A$ are: $\lambda = 3, -6, 21$
the eigenvalues of the matrix A^T are: $\lambda = 1, -2, 7$
the eigenvalues of the matrix A^2 are: $\lambda = 1, 4, 49$
the eigenvalues of the matrix A^{-1} are: $\lambda = 1, \frac{1}{-2}, \frac{1}{7}$

iii) The characteristic equation of A is $|A - \lambda I| = 0$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 4 & 3 \\ 0 & -2 - \lambda & 5 \\ 0 & 0 & 7 - \lambda \end{vmatrix} = 0 \quad \text{expand from the last row we get:}$$

$$= (1 - \lambda)\{(-2 - \lambda)(7 - \lambda) - 0\}$$

$$= (1 - \lambda)(-2 - \lambda)(7 - \lambda) = 0$$

.Expand this determinant to get the characteristic polynomial,

$$(1 - \lambda)(-2 - \lambda)(7 - \lambda) = \lambda^3 - 6\lambda^2 - 9\lambda + 14 = 0$$

Cayley-Hamilton theorem states that; a square matrix satisfies its characteristic equation.

Therefore, we must show that, $A^3 - 6A^2 - 9A + 14I = 0$

$$A^3 - 6A^2 - 9A + 14I = \begin{bmatrix} 1 & 12 & 291 \\ 0 & -8 & 195 \\ 0 & 0 & 343 \end{bmatrix} - 6 \begin{bmatrix} 1 & -4 & 44 \\ 0 & 4 & 25 \\ 0 & 0 & 49 \end{bmatrix} - 9 \begin{bmatrix} 1 & 4 & 3 \\ 0 & -2 & 5 \\ 0 & 0 & 7 \end{bmatrix} + 14 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

iii) From Cayley-Hamilton theorem, we know that;

$$A^3 - 6A^2 - 9A + 14I = 0 \quad \Rightarrow \quad -14I = A^3 - 6A^2 - 9A + 14I$$

$$I = -\frac{1}{14}(A^3 - 6A^2 - 9A) = -\frac{1}{14}(A^2 - 6A - 9I).A = A^{-1}.A$$

$$\text{Therefore, } A^{-1} = -\frac{1}{14}(A^2 - 6A - 9I)$$

$$A^{-1} = -\frac{1}{14} \begin{bmatrix} -14 & -28 & 26 \\ 0 & 7 & -5 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -13/7 \\ 0 & -1/2 & 5/14 \\ 0 & 0 & 1/7 \end{bmatrix}$$

A. Prof. Elsayed Mahdy

AI – 101 Mathematical Foundations and Software Development for AI

Matrix Diagonalization

Matrix diagonalization is the process of taking a *square matrix* and converting it into a special type of matrix--a so-called *diagonal matrix*.

Definition: An $n \times n$ matrix A is **diagonalizable** if there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix could be written in the form:

$$D = P^{-1}AP$$

where P is a matrix composed of the eigenvectors of A , D is the diagonal matrix constructed from the corresponding eigenvalues, and P^{-1} is the matrix inverse of P .

Theorem (1): An $n \times n$ matrix A is **diagonalizable** if and only if it has n linearly independent eigenvectors.

Theorem (2): An $n \times n$ matrix A has n distinct eigenvalues, then the corresponding eigenvectors are linearly independent and A is **diagonalizable**.

Matrices that are not diagonalizable

Some matrices are not diagonalizable over any field, most notably nonzero nilpotent matrices. For example, consider the following matrices;

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & -4 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

These matrices are *not diagonalizable* because there is no matrix P such that PAP^{-1} (or PBP^{-1} or PCP^{-1} or PDP^{-1}) is a diagonal matrix. Indeed, A , B , C and D are *nilpotent matrices*.

How to diagonalize a matrix

Example: Consider a matrix, $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & 2 \end{bmatrix}$. This matrix has eigenvalues: $\lambda = 3, 2, 1$

A is a 3×3 matrix with 3 different eigenvalues; therefore, it is diagonalizable. Note that if there are exactly n distinct eigenvalues in an $n \times n$ matrix then this matrix is diagonalizable(**Theorem (2)**)

These eigenvalues are the values that will appear in D the diagonalized form of matrix A , so by finding the eigenvalues of A we have diagonalized it. We could stop here, but it is a good check to use the eigenvectors to diagonalize A .

The corresponding eigenvectors of A respectively are ;

$$X_{\lambda=3} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}, \quad X_{\lambda=2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad X_{\lambda=1} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}.$$

Let P be the matrix with these eigenvectors as its columns, then $P = \begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix}$, then we can

write; $D = P^{-1}AP$

Remark: Note that there is no preferred order of the eigenvectors in P ; changing the order of the eigenvectors in P just changes the order of the eigenvalues in the diagonalized form of A .

Then P diagonalizes A , as a simple computation confirms, having calculated P^{-1}

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = P^{-1}AP = \begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & 2 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

Application: Diagonalization can be used to compute the powers of a matrix A efficiently, provided the matrix is diagonalizable.

$$\therefore D = P^{-1}AP \quad \Leftrightarrow \quad A = PDP^{-1}$$

$$(1) \quad \therefore A^k = P(D^k)P^{-1}$$

$$(2) \quad \therefore e^A = P(e^D)P^{-1}$$

Example: From the last example, since

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 0 & -1 & 0 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore,

$$(1) \quad A^k = P(D^k)P^{-1} = \begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 0 & 1^k \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$(2) \quad e^A = P(e^D)P^{-1} = \begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} e^3 & 0 & 0 \\ 0 & e^2 & 0 \\ 0 & 0 & e^1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

THANK *you*